

Helena Marsh

*Associate Professor of Materials
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Research overview. I direct the Ionic Interfaces Laboratory at Meridian University, where my group studies charge transport at grain boundaries and electrode–electrolyte interfaces in solid-state battery materials. Our work combines synthesis, operando spectroscopy, and machine-learning-guided screening to design layered-oxide cathodes and ceramic electrolytes with improved cycling stability. I teach undergraduate and graduate courses in materials thermodynamics and electrochemical energy storage, and I have advised eleven graduate students to degree completion.

Education

- 2007–2011 **Ph.D., Materials Science and Engineering**, *Vertex Institute of Technology*, Ashford, ME
Dissertation: *Ionic Transport Mechanisms in Layered Oxide Cathodes for Solid-State Batteries*. Advisor: Prof. Daniel Okafor.
- 2005–2007 **M.S., Materials Science and Engineering**, *Vertex Institute of Technology*, Ashford, ME
Thesis: *Sol-Gel Synthesis of Nickel-Rich Cathode Precursors*.
- 2001–2005 **B.S., Chemistry, magna cum laude**, *Apex University*, Rowanfield, NJ
Minor in Physics. Departmental Honors Thesis on lithium intercalation kinetics.

Academic Appointments

- 2021–present **Associate Professor (tenured)**, *Department of Materials Science and Engineering, Meridian University*, Ashford, ME
- 2015–2021 **Assistant Professor**, *Department of Materials Science and Engineering, Meridian University*, Ashford, ME
- 2012–2015 **Postdoctoral Research Associate**, *Nimbus College, Solid-State Chemistry Group*, Redgate, MA
Advisor: Prof. Sarah Lindqvist. Developed operando X-ray diffraction protocols for tracking phase transitions in cathode materials during cycling.
- 2011–2012 **Visiting Research Fellow**, *Synapse Foundation Materials Institute*, Corwin City, IL
Collaborative appointment supporting transition from doctoral to postdoctoral research.

Honors and Awards

- 2024 Distinguished Teaching Award, Meridian University
- 2022 Early-Career Faculty Fellowship, Synapse Foundation
- 2019 Young Investigator Award, Nexa Research Council
- 2017 CAREER Award, Nexa Research Council (Directorate for Materials Research)
- 2016 Rising Star in Materials Chemistry, Materials Research Society
- 2013 Postdoctoral Excellence Prize, Nimbus College
- 2011 Outstanding Doctoral Dissertation Award, Vertex Institute of Technology
- 2005 Chemistry Department Medal, Apex University

Research Funding